Yingying Liu

Department of Natural Environment, the University of Tokyo
(+81)080-4655-0825
email: 4964646891@edu.k.u-tokyo.ac.jp

EDUCATION

Natural Environment, The University of Tokyo **2023.10-Present**

Master of Science in Natural Environment

GPA: 3.9/4.0

Supervisor: Kei Yoshimura (Global Hydrology Group in Utokyo)

Atmospheric Science, Sun Yat-sen University **2019.09-2023.07**

Bachelor of Science in Atmospheric Science

GPA: 3.89/4.0 **Rank: 2/77**

Supervisor: Hua Yuan (Yongjiu Dai's land surface model group in SYSU)

PUBLICATIONS

- Ma, Q., **Liu, Y.**, Qiu, T., Huang, T., Deng, T., Hu, Z., Cui, T. Satellite-Observed Four-Dimensional Spatiotemporal Characteristics of Maritime Aerosol Types over the Coastal Waters of the Guangdong–Hong Kong–Macao Greater Bay Area and the Northern South China Sea (2022). *Remote Sensing*, *14*, 5464. https://doi.org/10.3390/rs14215464 **(Joint first authors)**.
- **Liu, Y.** and Yamazaki D. Evaluation of the Water Surface Elevation Measurement from SWOT Satellite in Small Rivers across the North Kyushu Region (2024). *Journal of Hydrology: Regional Studies*. (in review)
- Liu, Z., Yuan, H., Dong, W., **Liu, Y.**, Zhang, Y., Li, X., Xiang, J., Lin, W., Shi, J., & Dai, Y. (2024). Assessing daytime discrepancies and key factors in urban thermal environments: A local climate zones-based modeling study in five Chinese cities. *Urban Climate*, 55, 101993. https://doi.org/10.1016/j.uclim.2024.101993
- **Liu, Y.**, Yuan, H., Dong W., Lin, W., Liu, Z., Xiang, J., Land use and land cover change significantly shifted local water availability over the past three decades. (to be submitted)

RESEARCH EXPERIENCE

Development of an Online Data Assimilation Module for a Large-scale Hydrological Model CaMa-Flood **(2023.10-Present)**

Supervisor: Prof. Kei Yoshimura

- Design methods to generate effective ensemble simulations in a hydrological model.
- Developed an online data assimilation module for the CaMa-Flood Model.
- Applied water discharge observation to assess the improvements of online data assimilation module during extreme flood events.

Evaluation of the Water Surface Elevation Measurement from SWOT Satellite in Small Rivers across Japan **(2024.04-Present)**

Supervisor: Assoc. Prof. Dai Yamazaki

- Evaluated the performance of SWOT Raster data in small rivers to identify conditions under which it provides accurate measurements.

- Develop effective filters for SWOT Raster data to enhance measurement quality while maintaining sufficient data coverage.

***Impacts of land use and land cover changes on local water availability* (2022.03-2023.09)**

Supervisor: Assoc. Prof. Yuan Hua (Outstanding Bachelor Thesis)

- Analyzed the global land use and land cover change conditions using multi-source datasets.
- Applied a space-for-time approximation method to assess the impacts of land use and land cover changes on local water circulation.
- Identified mechanisms affecting water availability and enhanced understanding of meso-scale vegetation-climate feedback.

AWARDS

China National Scholarship **(top 0.2%)**	2022
The 9th UTEC Young Researcher's Challenge Support Program in Japan **(top 0.7%)**	2024
Outstanding Bachelor Theses Award **(top 5%)**	2023
Grand Prize of the 17th Challenge Cup in Guangdong province **(top 5%)**	2023
First Prize in Shenzhen Mathematical Modeling Competition **(ranked 1/1160)**	2021
First Prize Scholarship in Sun Yat-sen University **(ranked 2/106)**	2022
Science Competition Award in Sun Yat-sen University **(top 2%)**	2022

FUTURE RESEARCH PLANS

- Develop new product to provide wider coverage of dam water level and discharge with SWOT satellite.
- Develop an advanced dam operation simulation scheme for a large-scale hydrological model, incorporating insights from model comparisons and a new satellite-derived dam observation product.
- Evaluate the impacts of human activities, such as dam operations and reservoir management, on global hydrology through the advanced dam operation scheme.

ACADEMIC CONFERENCES

- Poster: Enhancing Data Assimilation Performance in Hydrological Models Through More Effective Ensemble Simulations: A Case Study during Typhoon Hagibis in Japan. **(AGU2024)**
- Poster: Evaluate the performance of SWOT measurement in small rivers. **(AGU2024)**

RELEVANT RESEARCH SKILLS

- Programming languages: Python, MATLAB, Fortran, and NCL.
- Proficient in numerical model: ILS and CoLM (land surface models), CaMa-Flood (hydrodynamic model), and WRF (meso-scale metrological model).
- Skilled in using Linux operating system.